Improving Diagnostic Colonoscopy Completion After Positive Stool-Based Screening in Health-Disparity Populations: A Systematic Review and Meta-analysis

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ABSTRACT

BACKGROUND: Delays in diagnostic colonoscopy after abnormal stool-based colorectal cancer screening are common in health-disparity populations and are associated with more advanced-stage disease at diagnosis. We evaluated the effectiveness of interventions designed to increase completion of diagnostic colonoscopy after abnormal fecal immunochemical test (FIT) or fecal occult blood test (FOBT) results in these populations.

METHODS: We searched six databases from inception through September 23, 2024 (updated June 4, 2025), for randomized trials, nonrandomized controlled studies, and controlled pre–post studies of interventions to improve colonoscopy completion after abnormal FIT or FOBT in health-disparity populations, as defined by the U.S. Department of Health and Human Services. Risk of bias was assessed with RoB 2 and ROBINS-I and certainty with GRADE. Studies were pooled within intervention types (PROSPERO CRD42024591109).

RESULTS: Ten studies met inclusion criteria: patient navigation (8), a registry-based quality-improvement collaborative (1), and low-touch result notification (1). In the registered primary analysis—navigation, with completion over each study's own window—navigation increased completion versus usual care (risk ratio [RR], 1.27; 95% CI, 1.10–1.46; moderate certainty). The effect was stable when all interventions were pooled as registered (RR, 1.28; 95% CI, 1.12–1.46) and across analytic choices, and was consistent in post hoc analyses at 6 months (RR, 1.29; 95% CI, 0.97–1.71) and 12 months (RR, 1.34; 95% CI, 1.20–1.49; $I^2 = 0\%$). Pooled effects did not differ between randomized and nonrandomized studies. The other two strategies rested on single studies; detection data were sparse.

DISCUSSION: Patient navigation improves timely diagnostic colonoscopy completion after abnormal stool-based screening in health-disparity populations (moderate certainty); other strategies increased completion but rest on single studies. Limitations include intervention heterogeneity, varying follow-up intervals, and few studies per intervention type. Standardized time-based follow-up benchmarks and expanded navigation may advance equity. This study received no external funding.

KEY WORDS: mass screening; colorectal neoplasms; colonoscopy; systematic review; meta-analysis

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INTRODUCTION

Colorectal cancer (CRC) remains a leading cause of cancer-related mortality in the United States.[1] Although CRC incidence and mortality have declined substantially among adults aged 50 years and older, largely because of increased screening, these gains have not been equitably realized.[2,3] Persistent racial, ethnic, and socioeconomic disparities remain, with non-Hispanic Black Americans experiencing the highest incidence and mortality and underserved populations consistently demonstrating lower screening uptake and poorer follow-up after abnormal screening results across multiple care settings.[4]

Stool-based screening is a two-step process: completing the screening test and, when it is positive, completing diagnostic colonoscopy. These are distinct steps, and much of the existing evidence and quality infrastructure has centered on the first. Screening participation approaches 74%, yet diagnostic follow-up is measured less often and appears substantially lower, particularly among socioeconomically disadvantaged populations.[5,6] In safety-net health systems, fewer than half of patients complete diagnostic colonoscopy within 1 year of a positive fecal immunochemical test (FIT).[7] Because a positive stool test confers no benefit unless it leads to colonoscopy, follow-up completion is the metric most proximate to cancer outcomes. Delays beyond 9–10 months are associated with

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increased colorectal cancer detection and more advanced stage at diagnosis.[8]

Barriers to timely diagnostic colonoscopy after abnormal CRC screening operate at both patient and system levels. Patient-facing barriers include limited health literacy, language discordance, transportation challenges, and financial constraints.[9] System-level factors, such as fragmented care coordination, inadequate referral tracking, and lack of standardized workflows for abnormal test follow-up, further contribute to delays.[10]

These challenges are compounded by the absence of a clearly defined, enforceable benchmark for timely diagnostic follow-up after abnormal stool-based screening.[11] Although professional organizations, including the U.S. Preventive Services Task Force and the American Cancer Society, emphasize the importance of prompt diagnostic evaluation, no guideline specifies a definitive timeframe linked to quality metrics or accountability. As a result, follow-up practices vary widely across health systems, and disparities in colonoscopy completion persist.[12] Such a benchmark is not without precedent: breast cancer screening programs, including the Centers for Disease Control and Prevention's National Breast and Cervical Cancer Early Detection Program, use a 60-day diagnostic-resolution standard, which surfaces the same racial and rural disparities seen in colorectal cancer follow-up.[13] An analogous standard therefore appears feasible for stool-based screening.

In response, health systems have implemented interventions such as patient navigation, electronic health record–based tracking and reminder systems, team-based accountability models, and culturally tailored outreach programs.[14] Prior systematic reviews have addressed related questions but mostly pooled screening uptake with diagnostic follow-up, studied general rather than underserved populations,[15] or synthesized evidence narratively without effect sizes or certainty grading.[16–18] Most directly, Issaka and colleagues descriptively reviewed follow-up of abnormal stool tests in safety-net settings and found navigation promising but the evidence limited and heterogeneous.[19] A quantitative synthesis restricted to health-disparity populations that distinguishes intervention types and grades certainty is therefore needed.

Accordingly, we conducted a systematic review to synthesize evidence on interventions designed to improve completion of diagnostic colonoscopy among health-disparity populations following abnormal FIT or fecal occult blood test (FOBT) results. We aimed to evaluate the effectiveness of these interventions and describe implementation features relevant to health disparity settings.

METHODS

Data Sources and Searches

We conducted this systematic review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Supplement Material 1).[20] We prospectively registered the study protocol in PROSPERO (CRD42024591109).

A medical librarian searched MEDLINE, EMBASE, CENTRAL, CINAHL, Scopus, and Web of Science from inception through September 23, 2024, combining controlled vocabulary and keywords for fecal-based screening, abnormal test follow-up, colonoscopy, and underserved populations, limited to English (Supplement Material 2). The search was updated June 4, 2025 (Supplement Material 3).

Eligibility Criteria

We included peer-reviewed randomized trials, nonrandomized controlled trials, and controlled pre–post studies comparing an intervention with usual care for increasing diagnostic colonoscopy completion in adults after a positive FIT or FOBT. Usual care was the study institution's standard follow-up process without navigation, outreach, or workflow modification.

Eligible populations were health-disparity populations as defined by the U.S. Department of Health and Human Services, groups facing persistent barriers related to race or ethnicity, language, socioeconomic status, insurance, disability, or documented discrimination.[4] Studies qualified if they explicitly targeted such populations or were conducted in settings predominantly serving them (federally qualified health centers, safety-net systems, community health centers). We excluded conference abstracts, gray literature, and duplicate publications without new data.

Study Selection

Six reviewers in three independent pairs screened titles, abstracts, and full texts in duplicate using Covidence, after a structured calibration process (three rounds of ten abstracts per pair, repeated until Cohen's kappa reached ≥ 0.80, and repeated at full-text screening). Disagreements were resolved by discussion or third-reviewer adjudication.

Data Collection and Extraction

Using a piloted, standardized form, reviewers extracted study design, year, country, setting, eligibility criteria, participant demographics (age, sex, race or ethnicity, primary language, insurance), and outcomes.

Reviewers characterized interventions using the Template for Intervention Description and Replication (TIDieR) checklist.[21] We classified each intervention by primary mechanism, such as patient navigation, other patient-directed strategies (mailed outreach, incentives, low-touch notification), provider-directed strategies, and system-level strategies such as registry-based tracking with audit and feedback. Studies combining navigation with a system component were

labeled navigation plus tracking. Two reviewers classified independently, since the label determined how studies were grouped.

Outcomes of Interest

The primary outcome was diagnostic colonoscopy completion within each study's prespecified timeframe. Secondary outcomes were time to colonoscopy; completion within 6 to 9 months of a positive stool test;[22,23] and colonoscopy findings, including detection of colorectal cancer or advanced adenomas.[14,22,24,25]

Data Synthesis and Analysis

Study selection is shown in a PRISMA 2020 flow diagram (Fig. 1). We prioritized intention-to-treat estimates for the primary outcome within each study's prespecified timeframe. Consistent with prior literature,[22,26] we treated FIT and FOBT as equivalent fecal-based modalities and recorded index test type for sensitivity analyses. Results are displayed in structured tables and forest plots.

When effect estimates were not reported directly, we calculated them from available summary data. Given the binary outcome and expected clinical heterogeneity, we synthesized results as risk ratios.

We conducted meta-analyses using the *meta* package in R (version 4.4.0). The registered primary outcome was completion of diagnostic colonoscopy within each study's own prespecified follow-up window, and the primary analysis pooled studies at that window. The primary analysis was restricted to patient navigation, the only intervention type evaluated by more than one study; the remaining interventions were pooled secondarily. We additionally examined completion at 6 and 12 months, intervals beyond which delayed colonoscopy after an abnormal stool test has been associated with higher colorectal cancer incidence and more advanced stage at diagnosis.[8] For comparisons including more than three studies, we applied inverse-variance random-effects models using restricted maximum likelihood estimation with Hartung–Knapp adjustment to account for between-study heterogeneity and improve robustness of inference. For comparisons including three or fewer studies, we applied fixed-effect inverse-variance models. Given the few studies, we compared estimates from low- versus higher-risk-of-bias studies informally rather than by meta-regression.

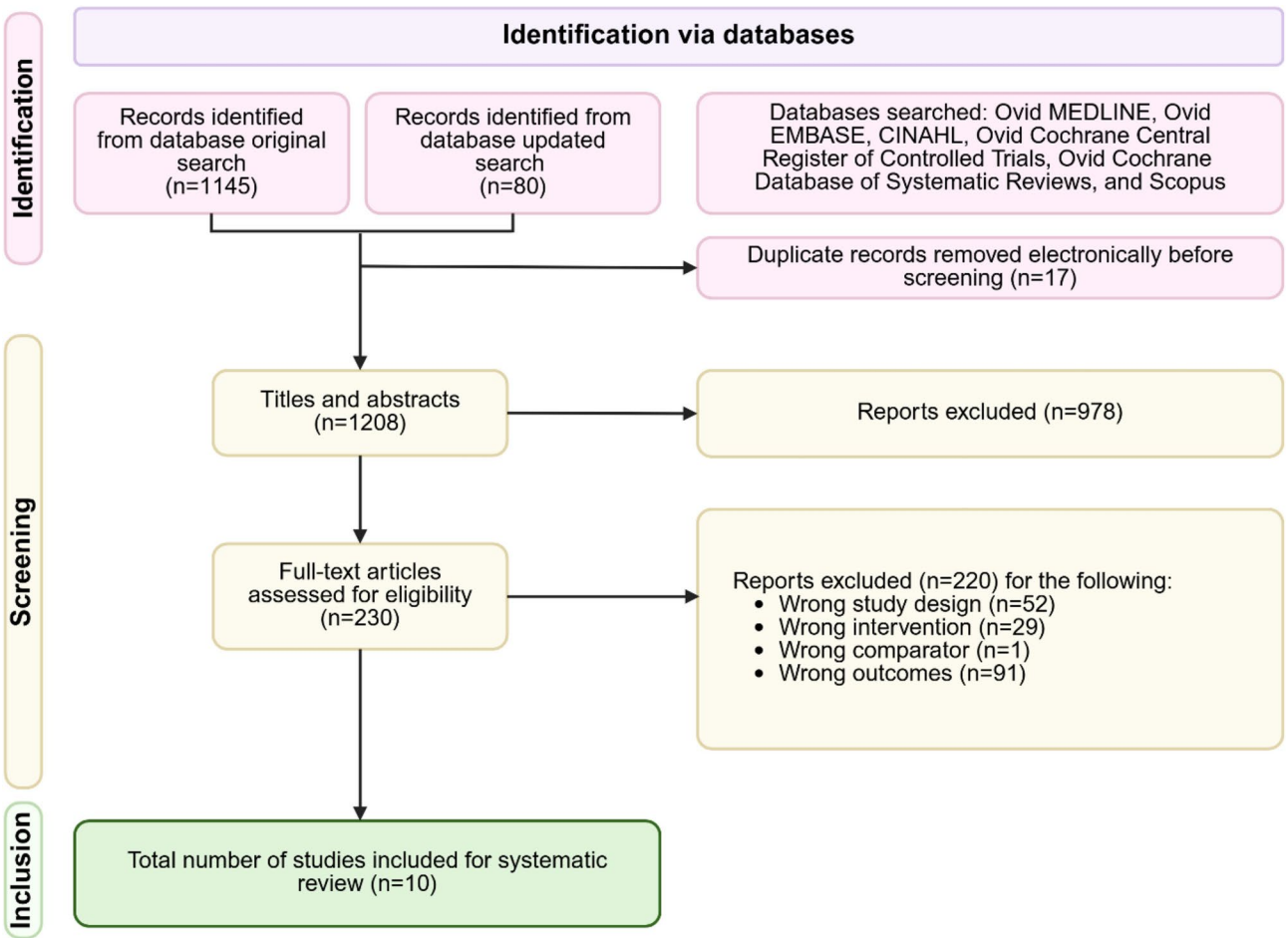


Figure 1 PRISMA flow diagram of study selection.

Risk of Bias Assessment

Two reviewers independently assessed risk of bias for each included study. Reviewers used the Cochrane Risk of Bias 2 (RoB 2) tool[27] for randomized controlled trials and the Risk of Bias in Non-Randomized Studies of Interventions (ROBINS-I)[28] tool for nonrandomized studies. Reviewers resolved disagreements through discussion. Judgments followed each instrument's guidance and informed interpretation and certainty ratings.

Certainty of Evidence

Two reviewers independently graded certainty of evidence for each outcome using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework. Reviewers assessed five domains: risk of bias, inconsistency, indirectness, imprecision, and publication bias.

Because risk of bias in nonrandomized studies was assessed with ROBINS-I, which evaluates confounding and selection bias directly, all studies began at high certainty; the certainty of nonrandomized evidence was then reflected through the risk-of-bias domain rather than by a lower starting rating, avoiding double-counting. Certainty was downgraded for serious risk of bias, inconsistency, indirectness, imprecision, or suspected publication bias, and could be upgraded for a large effect magnitude or when residual confounding would bias toward the null. Domain-specific judgments appear in Supplement Table 9.

Role of the Funding Source

No funding source was required for this study.

RESULTS

Study Selection

After removing 15 duplicates, 1130 unique records were screened and 225 full texts assessed ($\kappa = 0.90$); 218 were excluded, most often for ineligible outcomes ($n = 91$), incorrect design ($n = 51$), or out-of-scope interventions ($n = 29$). Seven studies met eligibility criteria; an updated search identified three additional studies, yielding ten included studies (Fig. 1).

Study Characteristics and Participants

Seven studies were randomized controlled trials,[14,22,24,26,29,30] one was a prospective cohort study,[25,31,32] and two were pre–post study designs.[32,33] Eight studies evaluated follow-up after a positive FIT;[14,22,24–26,29,31,32] one focused on FOBT[33], and one included multiple abnormal screening modalities.[30]

Across studies, 73,629 adults were eligible for screening, of whom 60,845 (82.6%) completed stool-based screening, and 18,478 (30.4%) had abnormal results. Participant characteristics and study features are summarized in Table 1.

Included studies were conducted in settings predominantly serving racial and ethnic minority, low-income, and limited-English-proficiency patients: federally qualified health centers, integrated and county safety-net systems, and a Veterans Affairs system. Several trials enrolled largely Hispanic/Latino populations with substantial proportions of Spanish-speaking participants and used bilingual navigators or translated materials; two large safety-net trials enrolled predominantly uninsured Black and Hispanic adults; one enrolled rural Medicaid and dual-eligible enrollees; and one studied older male veterans. Reporting of race, ethnicity, language, and rurality was inconsistent,

Table 1 Characteristics of Included Studies

Study (year, country); reference no	Study design	Setting*	Population	Intervention $N^{\dagger}$	Control $N^{\dagger}$
Coronado (2025, USA)[14]	RCT	FQHC	Adults 50–75 with abnormal FIT	479	488
Coronado (2025, USA)[22]	Cluster RCT	Rural clinics	Medicaid/dual-eligible adults 50–75	2613	3001
Coronado (2021, USA)[29]	RCT	Integrated system	Adults 50–75 with abnormal FIT	200	215
Gupta (2016, USA)[24]	RCT	Safety-net	Uninsured adults 50–64	2000	6565
Idos (2021, USA)[25]	Prospective cohort	Safety-net	Adults 50–75 with abnormal FIT	1317	1214
Powell (2011, USA)[33]	Pre–post QI	VA system	Adults with positive FOBT	10,808	1971
Raich (2012, USA)[30]	RCT	Safety-net	Medically underserved adults	548	538
Reuland (2024, USA)[31]	RCT	FQHC	Adults 50–75 with abnormal FIT	600	194
Shareef (2024, USA)[32]	Pre–post QI	FQHC	Adults 50–75 with abnormal FIT	208	156
Singal (2017, USA)[26]	Pragmatic RCT	Safety-net	Uninsured adults 50–64	2400	3599

Abbreviations: CRC colorectal cancer, *FIT* fecal immunochemical test, *FOBT* fecal occult blood test, *FQHC* federally qualified health center, *QI* quality improvement, *RCT* randomized controlled trial, *VA* Veterans Affairs

*Settings served health-disparity populations as defined by the U.S. Department of Health and Human Services

†Intervention N and Control N are the enrolled/randomized counts reported by each study, not the abnormal-FIT population analyzed for colonoscopy completion ($n = 4612$)

precluding subgroup meta-analysis by demographic group. Despite a broad search spanning race, rurality, language, disability, housing, and insurance, we identified no eligible studies conducted specifically among patients with disabilities, people experiencing homelessness, or refugee and recent-immigrant populations.

Because few studies were available, individual features merit comparison. The Coronado group contributed four related trials showing how navigation has been adapted across settings and intensities: precision navigation in an integrated system, navigation in a largely Latino federally qualified health center, and mailed outreach plus navigation among rural Medicaid enrollees. Interventions succeeding among limited-English-proficiency patients shared bilingual, culturally concordant navigation and translated result letters, whereas the intervention tested among rural veterans relied on registry-based tracking with audit and feedback rather than individual navigation. For programs targeting specific subpopulations, navigation intensity and language concordance should be matched to the population served.

Intervention Characteristics

Seven studies tested patient-directed interventions,[14,22,24,26,29–31] one evaluated a system-level intervention (registry-based tracking with audit and feedback),[33] and two combined patient navigation with system-level tracking.[25,32] Follow-up intervals ranged from 60 days to 12 months, with one study not specifying a time benchmark.[24]

By mechanism, the interventions fell into three groups that cannot be implemented interchangeably. Patient navigation was the dominant strategy, evaluated in eight studies (seven as the primary intervention, one combined with incentives added to mailed outreach). The other two studies each tested a distinct approach: a registry-based quality-improvement collaborative with audit and feedback, and a low-touch mailed result-notification trial. Because these embody different theories of action, we synthesized them separately and treated navigation as the primary analysis.

Navigation typically involved outreach by trained bilingual lay or nurse navigators and included results notification, education, scheduling assistance, reminders, and follow-up.[14,22,25,30–32] Intervention characteristics are summarized in Table 2. Full intervention descriptions are provided in Supplement Table 2.

Colonoscopy Completion

Outcome definitions and follow-up intervals varied (Table 3). Most included studies assessed colonoscopy completion within 6 months of the abnormal test; three trials reported completion at 12 months (one

Table 2 Intervention Characteristics

Study (year, country); reference no	Intervention	Level	Delivered by	Core components	Duration/intensity
Coronado (2025, USA)[14]	PRECISE navigation	Patient	Bilingual lay navigators	Barrier assessment, education, scheduling, reminders, follow-up	Initial contact ≤ 1 week; follow-up as needed
Coronado (2025, USA)[22]	SMARTER CRC navigation	Patient	Trained navigators	Barrier assessment, prep education, reminders	Not reported
Coronado (2021, USA)[29]	Navigation	Patient	Nurse navigator	Education, scheduling, reminders, post-procedure follow-up	≤ 180 days
Gupta (2016, USA)[24]	Incentives + outreach	Patient	Nurse manager, assistants	Mailed FIT outreach, incentives, scheduling support	Colonoscopy ≤ 12 weeks
Idos (2021, USA)[25]	Navigation + tracking	Multicomponent	Nurse practitioners, physicians	Education, scheduling, reminders, missed-visit follow-up	Target ≤ 6 months
Powell (2011, USA)[33]	C4 QI collaborative	System	Multidisciplinary QI teams	Registry tracking, workflow redesign, audit-feedback	6–15 months
Raich (2012, USA)[30]	PNRP navigation	Patient	Trained navigators	Barrier assessment, care coordination, follow-up	Until resolution
Reuland (2024, USA)[31]	SCORE navigation	Patient	Bilingual navigators	Tailored outreach, scheduling, reminders	≥ 5 contacts
Shareef (2024, USA)[32]	Navigation + checklist	Multicomponent	Experienced navigators	Checklist-guided follow-up, barrier resolution	2 PDSA cycles
Singal (2017, USA)[26]	Result notification	Patient	Research staff	FIT result notification and referral	≤ 1 week

Abbreviations: CRC colorectal cancer, *FIT* fecal immunochemical test, *PDSA*, plan–do–study–act, *QI* quality improvement

cluster-randomized trial reported both windows), and two evaluated shorter windows (≤ 12 weeks and 60 days).

Across the seven navigation studies contributing to the primary analysis (4612 participants), 2165 patients completed diagnostic colonoscopy (1221 with navigation, 944 with usual care). The financial-incentive trial and the system-level quality-improvement initiative did not report arm-specific counts suitable for pooling and were summarized narratively (Supplement Table 3).

Among studies reporting pre-intervention or usual-care rates, colonoscopy completion ranged from approximately 21 to 50% at baseline and from 38 to 75% after intervention (Supplement Table 4).

Most randomized trials showed higher completion with patient-directed interventions than with usual care. The single trial of financial incentives found higher completion that was not statistically significant.[24] Nonrandomized studies similarly reported improved completion following quality-improvement or system-level interventions.[25,32,33] Some studies reported adjusted analyses accounting for demographic or system-level covariates,[22,25,30,33] whereas others presented unadjusted comparisons;[24,32] adjustment methods and covariates are summarized in Supplement Table 5.

Adenoma and Colorectal Cancer Detection

Few studies reported colonoscopy findings by intervention group. Available data showed no significant differences in adenoma or colorectal cancer detection between intervention and control groups.[24,25,31] Among participants completing colonoscopy after abnormal FIT, adenomas were detected in 45%, advanced neoplasia in 7%, and colorectal cancer in 3%. Study-specific findings are summarized in Supplement Table 6.

Time to Colonoscopy

Time from abnormal screening to colonoscopy ranged from 60 to 256 days, and studies reporting arm-specific outcomes consistently showed shorter intervals with intervention. Navigation reduced time to colonoscopy by approximately two to three weeks, whereas system-level QI initiatives achieved larger absolute reductions. Detailed time-to-colonoscopy outcomes are provided in Supplement Table 7.

Effect of Patient Navigation on Colonoscopy Completion

Patient navigation was the only strategy evaluated by more than one study and was therefore the primary analysis; completion was measured over each study's own follow-up window, as registered. Across seven studies,[14,22,25,29–32] navigation increased the likelihood of completing diagnostic colonoscopy by 27% compared with usual care (RR, 1.27 [95% CI, 1.10 to 1.46]; $I^2 = 52\%$; Fig. 2A). Study-level effects were consistent in direction and larger in smaller trials of more intensive navigation.

This estimate was robust to how the analysis was constructed (Fig. 3). It was essentially unchanged when the low-touch reminder trial was added (RR, 1.28 [95% CI, 1.12 to 1.46]; Supplement Fig. 1), and it was similar under fixed-effect and standard random-effects models.

In post hoc analyses restricted to a common follow-up window, the effect of navigation was consistent (Fig. 2B, C). At 6 months (five studies), completion increased with navigation (RR, 1.29 [95% CI, 0.97 to 1.71]; $I^2 = 52\%$; Fig. 2B); this estimate did not reach statistical significance under the Hartung–Knapp variance estimator ($P = 0.07$), although it did under fixed-effect and standard random-effects

Table 3 Colonoscopy Completion Outcomes by Study

Study	Study design	Follow-up interval	Prespecified timeframe	Meta-analysis disposition
PRECISE[14]	RCT	12 months (primary outcome)	Yes	12-month stratum
Coronado 2021[29]	RCT	6 months	Yes	6-month stratum
SMARTER CRC[22]	Cluster RCT	6 and 12 months	Yes	6- and 12-month strata; post-randomization subgroup
Raich[30]	RCT	12 months (diagnostic resolution–censored)	Yes	12-month stratum
Reuland[31]	RCT	6 months	Yes	6-months stratum; post-randomization subgroup
Idos[25]	Prospective cohort	6 months	Yes	6-months stratum; nonrandomized
Shareef[32]	Pre–post QI	6 months	Yes	6-months stratum; nonrandomized
Singal[26]	RCT	6 months	Yes (secondary)	Non-navigation comparison; registered pool only
Powell[33]	Pre–post QI	12 months (60-day timeliness reported)	No	System-level reporting; narrative synthesis only
Gupta[24]	RCT	≤ 12 weeks	No	Navigation delivered to all arms; not poolable

Studies are grouped by meta-analysis disposition: randomized trials contributing to the pooled analysis, then nonrandomized studies contributing to the pooled analysis, then studies not included in the pooled analysis

All included studies were conducted in the United States

Abbreviations: QI quality improvement, *RCT* randomized controlled trial

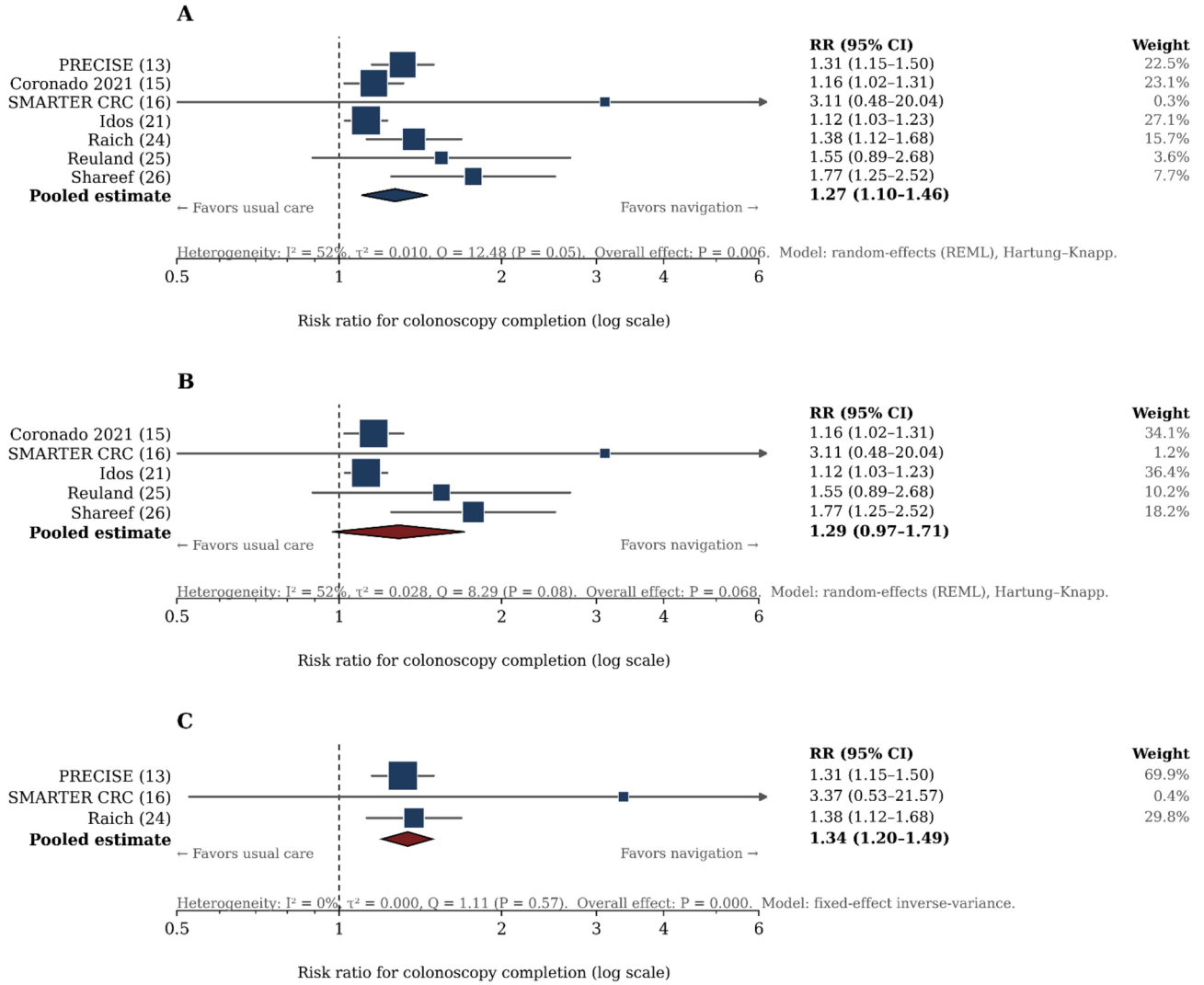
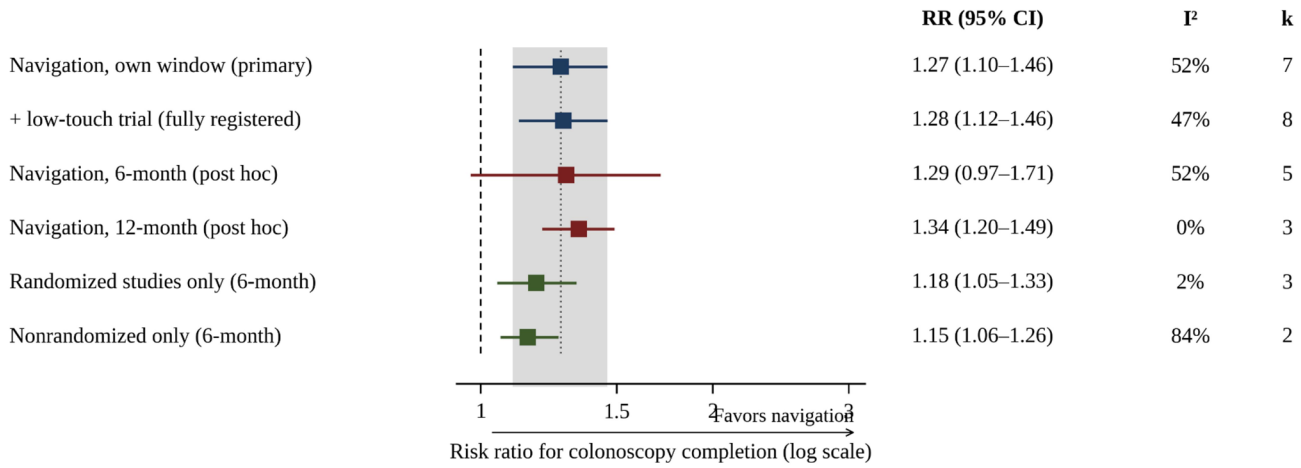


Figure 2 Forest plot (A–C) depicting the pooled effect of patient navigation on diagnostic colonoscopy completion after a positive stool-based screening test, relative to usual care. Squares denote study-specific risk ratios (area proportional to weight), horizontal lines the 95% confidence intervals, and diamonds the pooled estimates; the dashed line marks no effect (risk ratio = 1.0), with values to its right favoring navigation. Study-level risk ratios with 95% confidence intervals and weights appear in the right-hand columns, and heterogeneity and overall-effect statistics beneath each panel. Panel A shows the PROSPERO-registered primary analysis, in which completion is measured within each study's own prespecified follow-up window (random-effects model with Hartung–Knapp adjustment). Panels B and C are post hoc analyses that instead apply a common follow-up window to the prespecified completion outcomes: 6-month completion (panel B; random-effects model with Hartung–Knapp adjustment) and 12-month completion (panel C; fixed-effect inverse-variance model, as between-study variance cannot be reliably estimated from three studies).

models. At 12 months (three studies), navigation significantly increased completion (RR, 1.34 [95% CI, 1.20 to 1.49]) with no observed heterogeneity ($I^2 = 0\%$; Fig. 2C); the two largest of these trials randomized navigation among patients already known to have an abnormal test.

Risk of Bias Assessment

Risk-of-bias assessments are presented in Supplement Tables 8A and 8B. Of seven randomized trials, three were at low risk of bias and four raised some concerns (most often deviations from intended interventions or absent prespecified protocols); none was high risk. All three nonrandomized studies were at serious risk of bias, chiefly from confounding inherent to uncontrolled or pre–post designs. Across nonrandomized studies, outcome measurement was generally judged to be at low risk of bias because colonoscopy completion was assessed using objective electronic health record data. These assessments informed interpretation of pooled estimates and certainty-of-evidence ratings. Pooled estimates were similar across risk-of-bias strata in the 6-month window: RR 1.18 (95% CI, 1.05 to 1.33) among randomized studies and RR 1.15 (95% CI, 1.06 to 1.26) among nonrandomized studies. The largest single study-level estimate arose in a nonrandomized pre–post study, but nonrandomized studies did not yield systematically larger pooled effects. A sensitivity analysis restricted to the trials that randomized navigation among



Shaded band = 95% CI of the primary analysis; dotted line = its point estimate.

Figure 3 Forest plot of the pooled risk ratio for diagnostic colonoscopy completion with patient navigation across alternative analytic choices (sensitivity analysis). Each row is a separate pooled estimate: the primary navigation analysis; the same analysis with additional interventions added; the two window-stratified analyses; and the risk-of-bias strata. The shaded band marks the 95% confidence interval of the primary analysis and the dotted line its point estimate; the right-hand columns give each estimate's risk ratio with 95% confidence interval, I^2, and number of studies. Points to the right of the dashed no-effect line favor navigation.

abnormal-FIT patients yielded RR 1.33 (95% CI, 1.19 to 1.49; $I^2 = 0\%$), and excluding post-randomization subgroups did not materially change the estimate (Supplement Fig. 2).

Certainty of the Evidence

Certainty-of-evidence assessments are presented in Supplement Tables 9 and 10. Certainty was moderate for the primary navigation analysis and for 12-month completion, which showed consistent effects and no heterogeneity. Certainty for the 6-month estimate was low, reflecting imprecision. Evidence for time to colonoscopy was low, and certainty for adenoma and colorectal cancer detection was very low because of limited and inconsistently reported outcome data.

DISCUSSION

In this systematic review and meta-analysis, patient navigation was associated with clinically meaningful, moderate-certainty improvements in diagnostic colonoscopy completion after abnormal stool-based colorectal cancer screening in underserved settings. Completion rates rose from approximately 21–50% at baseline to 38–75% after intervention. These gains matter because delayed colonoscopy after a positive stool test is associated with more advanced stage at diagnosis.[8,34]

Navigation's benefit was directionally consistent across the randomized trials.[14,22,24,26,29,31] Its effectiveness likely reflects its ability to address multiple barriers simultaneously, including result communication, scheduling, transportation, insurance navigation, bowel preparation support, and reassurance.[16,17] These services were often delivered by bilingual navigators trained in motivational interviewing, an approach well suited to safety-net populations facing overlapping socioeconomic and structural barriers.[7,35] The two studies that layered registry-based tracking onto navigation were directionally consistent with this benefit but were few, nonrandomized, and at serious risk of bias, so they can only be regarded as hypothesis-generating regarding the added value of system-level components.

Interventions acted at different points along the diagnostic pathway: patient-directed strategies emphasized education, coordination, and reminders, whereas system-level tools added registries, standardized workflows, and performance monitoring.[24,26] The single system-level QI collaborative improved timing more than overall completion,[33] though even modest reductions in delay are clinically meaningful.[8,36] Taken together, the available evidence supports patient navigation as the best-supported strategy; whether adding system-level coordination yields further improvement remains an open question that the current studies cannot answer.[7,35]

Follow-up intervals varied widely, reflecting the absence of standardized quality benchmarks for diagnostic colonoscopy after abnormal stool-based screening.[5,23,37] No enforceable benchmark exists, contributing to practice variation, even though observational data consistently link each month of delay to higher colorectal cancer risk and more advanced stage.[8,36] Even modest reductions in delay may therefore confer meaningful benefit. The feasibility of such a benchmark is supported by experience in breast cancer screening, where a 60-day diagnostic-resolution standard has been operationalized, tied to reimbursement, and used to monitor equity; a comparable, evidence-based time target for colonoscopy after a positive stool test could similarly anchor accountability and disparity monitoring, provided it is paired with the navigation resources needed to meet it.[13]

Certainty was moderate for the primary navigation analysis and for 12-month completion (consistent, no heterogeneity), and low for the imprecise 6-month estimate and for detection outcomes. These findings support timely (6-month) improvements in follow-up completion. Because a completed colonoscopy is a discrete event rather than an ongoing treatment, the relevant question is whether navigation increases eventual completion or merely accelerates it. The concordance of the 6- and 12-month navigation estimates, and the absence of heterogeneity at 12 months, favor a true increase in completion rather than a shift in its timing.

Implementation context also mattered: geographic access, staffing, and referral infrastructure varied, and greater travel distance predicted lower likelihood of timely completion.[25,38] Effective interventions must therefore be adaptable to local workflows and resource constraints.[7,38,39]

Language and cultural concordance appeared important. Several studies used bilingual navigators[14,22,31] and culturally tailored materials, and evidence outside this review supports higher screening uptake with language concordance.[40,41] Given persistently lower screening rates among adults with limited English proficiency, culturally and linguistically concordant navigation is a plausible contributor to the benefits observed in the trials that enrolled these patients, although no included study formally tested language concordance as an independent component; this remains an important question for future research.

Clinically, the consistency of the navigation evidence supports embedding navigation in routine follow-up pathways after abnormal fecal-based screening in safety-net settings. The absence of standardized follow-up benchmarks remains a critical policy gap.[8,12,36,42] Establishing standardized, time-based quality metrics for colonoscopy after abnormal stool-based screening may represent an important step toward improving accountability and reducing disparities in follow-up care.

This review has several strengths, including comprehensive searches, rigorous dual-review processes, and systematic intervention characterization.[21,43] The principal limitations arise from the evidence base. Interventions and follow-up intervals were heterogeneous, and only one or two studies were available for every approach other than patient navigation, precluding formal comparison across intervention types. Rural populations were represented by a single trial, and reporting of participant demographics was inconsistent, precluding subgroup analysis. Adenoma and cancer detection were reported by few studies. All included studies were conducted in the United States, which may limit generalizability to other health systems.

Future research should define the optimal components and intensity of navigation, test whether system-level coordination adds benefit, evaluate scalability beyond grant-funded demonstrations, and extend evidence to populations absent from this review.

CONCLUSION

This systematic review suggests that patient navigation improves diagnostic colonoscopy completion after positive FIT/FOBT in health-disparity populations, whereas other strategies, financial incentives, low-touch reminders, and registry-based quality improvement show promise but rest on single studies. However, follow-up rates often remain below 75%, and meaningful variation persists across settings. Improving equity in colorectal cancer outcomes will likely require continued investment in patient-centered support, system-level care coordination, and clearer quality benchmarks that address structural barriers to follow-up.

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